Understanding Pizza Dough Hydration: A Science-Based Guide

Dough hydration is the ratio of water to flour used in a recipe, yet it impacts nearly every stage of the pizza-making process. From the stickiness of the dough on your workbench to the way the crust bakes in the oven, hydration is a key variable that bakers must learn to manage. This article provides a detailed look at what hydration is, how it affects dough handling, baking and crust characteristics, and provides a science-based perspective on common myths surrounding hydration in pizza dough

Defining and Calculating Dough Hydration

Before we explore how hydration impacts both the dough and the finished product, let's first define what hydration is and how to calculate it accurately.

What Is Hydration in Dough?

In baking, "hydration", also refers to the **amount of water in the dough**.

"Dough hydration" is also referred to as "dough absorption" or "baking absorption," particularly in professional or academic contexts.

Technically, hydration – like all other dough components – is calculated using baker's percentages. This means it is expressed as a **percentage** of the weight of **flour** in the dough.

Simply put, hydration is the **ratio of water to flour in the dough**.

Calculating Dough Hydration: An Example

Let's look at a simple example:

If a dough contains 100 grams of flour and 60 grams of water (or milliliters, since 1 ml of water weighs 1 gram), hydration is calculated by dividing the weight of the **water** by the weight of the **flour**:

60 ÷ 100 = 0.6

Thus, the hydration of this dough is **60%**.

Now consider another example: a dough with 180 grams of flour and 117 grams of water. Again, divide the weight of the water by the weight of the flour:

117 ÷ 180 = 0.65

In this case, the hydration of the dough is **65%**.

How Hydration Affects Dough Behavior and Handling (Before Baking)

In general, **higher** dough hydration – meaning a higher water content – results in a "wetter" dough. Such dough tends to be more **extensible** (stretchy), softer (as dough, not necessarily as a baked product), and stickier. As hydration increases beyond a certain point, the dough can become more challenging to work with.

On the other hand, **lower** hydration results in a drier, stiffer dough. It becomes more **elastic** (resistant to stretching), less sticky, and, within certain limits, easier to handle.

'Easy to work with' is a subjective term that varies from person to person. Generally, it refers to dough that doesn't stick excessively to your hands, stretch too much (overly extensible), or resist too much (overly elastic). This balance allows you to handle the dough effectively without compromising the final result, whether you're stretching it into a pizza base or shaping it for bread. What feels 'easy to work with' depends on individual experience and preferences.

Hydration Levels in Dough and Their Effects on Handling

Here's a general guide to dough hydration levels and their impact on the dough's behavior and handling:

- **0-45% – Very Low Hydration**: This results in a very stiff dough that requires significant physical effort to work with. Examples include pasta dough or shortcrust pastry (pie crust).
- **45-55% – Low Hydration**: This results in a relatively stiff dough, though it is easier to handle than very low hydration dough.
- **56-64% – Medium Hydration**: This is the typical range for most baked goods, resulting in dough that is easy to work with and not too sticky for most people.
- **65-74% – High Hydration**: This leads to a sticky, very extensible dough.
- **75% and Above – Very High Hydration**: This creates an extremely sticky and extensible dough.

For more on dough's elasticity and extensibility and their impact on dough, see the article Dough Elasticity and Extensibility: Understanding the Two Most Important Properties in Pizza Dough.

Additional Variables That Affect Dough Behavior

While the hydration level plays a crucial role, other factors also affect dough's behavior, including its extensibility, elasticity, and stickiness:

- **Flour Used**: Different flours have varying water absorption capacities, which directly affect dough behavior. Dough made from flour with a high water absorption capacity will feel less sticky than dough made from flour with lower absorption capacity, even if both have the same hydration level.
- **Dough Temperature**: Colder dough will always be more elastic, less extensible, and less sticky compared to dough at a higher temperature.
- **Dough Composition**: Adding fat to the dough increases its extensibility and contributes to a "wetter" feel (more on this later). Additionally, high amounts of sugar (above 5%) can make the dough feel stickier.
- **Kneading Method**: The kneading method used directly influences the dough's extensibility and elasticity.

The Relationship Between Hydration and Fermentation Rate

In general, the higher the water content in dough, the more **free water** is available. **Theoretically**, this "extra" water could be utilized by yeast and bacteria during fermentation, potentially increasing the rate of fermentation.

However, **in practice**, higher hydration does not necessarily result in a noticeable increase in fermentation rate.

While yeast activity can slightly increase with more free water in the dough, its impact diminishes once the flour absorbs sufficient water (typically around 58% hydration, depending on flour type). Beyond this point, additional water has only a **marginal** effect on yeast and bacteria activity. The excess water primarily alters the dough' texture, rather than significantly enhancing biological availability for yeast, bacteria and enzymes.

Simply put, dough at 80% hydration won't ferment faster than dough at 70% hydration. This principle applies to hydration levels above approximately 60%.

The misconception that higher hydration increases the fermentation rate stems from the visual appearance of high-hydration doughs. These doughs "puff up" more during fermentation compared to lower-hydration doughs. This effect is due to their increased extensibility, which allows them to stretch more easily, giving the illusion of faster fermentation because of their greater volume.
In conclusion: hydration is not one of the factors that determine the fermentation rate. While it influences dough characteristics, its impact on the speed of fermentation is minimal once the flour has absorbed sufficient water.

Should Oil (and Other Ingredients) Be Considered Part of Dough Hydration?

Hydration calculations often cause confusion, particularly about whether ingredients other than water should be included.

In general, the term "hydration" refers only to **water**, which is absorbed by the flour and **influences various processes in the dough**. Specifically:

- The water absorbed by the dough drives starch gelatinization – an essential process during baking that contributes to the structure and final texture of baked products.
- For gluten bonds to form, the flour **must** absorb **water**.

In the following sections, we will explore whether additional liquids (and ingredients) other than water should be considered part of dough hydration.

The Distinct Effects of Oil vs. Water

Because oil contains no water– being 100% fat – it does not contribute to the processes described above. Moreover, oil "coats" the flour particles, **preventing them from absorbing water**. This makes oil's effect on the dough entirely different from that of the water.

You can read more about the effects of oil on dough in the following article: Oil in Pizza Dough Explained: Why to Use It and How It Affects the Dough [Texture, Flavor & More].

Therefore, it is incorrect to include oil as part of the dough's hydration. Oil and water have distinct effects on the dough, and cannot be substituted.

However, oil does influence the **rheological** properties of the dough – **how it feels and handles**. As a liquid (even though it is not water), oil adds to the sensation of "wetness" in the dough, making it softer and more extensible.

Later, we will discuss the distinctions between moisture, wetness, and softness.

In conclusion, While oil makes the dough feel “wet,” it does not chemically interact with flour as water does, nor does it participate in the key biochemical processes driven by water. Therefore, oil **should not** be included when calculating dough hydration.

Accounting for Water in Milk, Eggs, and Other Ingredients

When calculating dough hydration, it is important to account for ingredients that contain varying amounts of water. For example:

- **Honey**: About 15% water
- **Butter**: About 16% water
- **Margarine**: About 20% water
- **Eggs**: About 75% water
- **Milk**: About 88% water

When these ingredients are added to the dough, they effectively **introduce additional water**. Depending on the quantity used, this can **significantly affect the dough’s overall hydration** (i.e., its actual water content).

Therefore, to calculate hydration accurately, it is essential to include the water contributed by these other ingredients.

An Example: Effect of Different Ingredients on Dough Hydration

Let’s consider the following enriched dough recipe for challah (for simplicity, without salt, sugar, and yeast):

Ingredient	Quantity (Grams)	Baker’s %	Water Content (Grams)	Hydration (Baker’s %)
Flour	200	100%	–	–
Water	100	50%	100	50%
Eggs	20	10%	15(20*0.75)	7.5%(15/200)
Butter	20	10%	3(20*0.16)	1.5%(3/200)

At first glance, it may seem that the hydration is **50%,** but this calculation is **incorrect**.

The eggs and butter contribute an **additional 18 grams** of water to the dough, which represents **9%** hydration in baker's percentages (7.5% from the eggs + 1.5% from the butter). Therefore, when including the water from the eggs and butter in the calculation, the **actual** hydration is **59%** – a significant difference.

Now consider a contrasting example involving milk, which contains about 88% water. If you replace the water in the recipe with milk, note that the amount of water, and consequently the actual hydration, will be lower.

Continuing with the previous example, replacing the water with milk results in hydration from the milk being 45% (88 grams of water) instead of 50% (100 grams of water). Again, this difference is significant.

The Effects of Hydration on Baking and the Final Product

For a complete understanding of the following sections and how hydration affects various processes in dough during baking, I highly recommend reading the article [From Dough to Pizza: A Deep Dive into the Processes Occurring Inside the Dough During Baking].

Dough hydration plays a key role in baking, impacting both the baking process and the final texture of the baked product. In the following sections, we will explore these effects.

Terminology for the Next Sections:

Crust: The outer layer of the baked product that dries, hardens, and browns during baking. For pizza, this includes the bottom and the exterior rim (cornicione). For bread, it refers to the entire exterior.
Crumb: The soft, inner part of the baked product. In pizza, this refers to the interior of the rim, while in bread, it's the entire inner section.

Open Crumb Structure: A crumb structure characterized by a more "open" and airy texture, with larger air cells or "bubbles".
Closed Crumb Structure: A crumb structure that is denser and less airy, with smaller air cells or bubbles.

Note that the above refer only to the **size and volume** of the air cells. The **quantity** of air pockets is determined solely during kneading, and does not change during fermentation and baking. Dough hydration does not affect the **number** of air pockets, only their final **volume**.

The General Effect of Hydration During Baking

This section explains the effects of **increased hydration** on dough during baking, and serves as a foundation for understanding the additional impacts of hydration, which will be explored in the following sections.

In general, the **higher** the dough hydration:

1. The dough becomes **softer and more extensible**, allowing it to stretch more during baking.
2. **More steam is generated during baking** due to the higher water content in the dough.
3. The increased steam creates **more pressure** within the dough. The greater pressure causes the gas bubbles (CO2) within the dough to expand.
4. The combination of a **more extensible dough** and **increased steam** causes the **air cells**, filled with millions of gas bubbles and surrounded by the gluten network, to **expand further**.
5. The result is a dough with **larger air cells** ("bubbles"), and **greater overall volume**.

In summary, higher hydration generates more steam during baking, which creates greater internal pressure within the dough. This pressure causes the gas bubbles to expand, pushing outward the walls of the air cells they occupy.

As a result, the air cells gain more volume, the gluten network surrounding them expands as well, and the final product has greater volume and larger air bubbles.

In simple terms: the **higher** the dough hydration, the **greater** the oven spring, the **more volume** the dough will gain during baking, and the **more open and airy** the crumb structure.

The Impact on Baking Mechanics and Thermodynamics

We now know that higher hydration results in larger air bubbles forming in the dough during baking, leading to a more open crumb structure. Now, let's dive deeper into how this impacts the **baking process**:

1. More efficient movement of steam within the dough: A more open crumb structure allows steam generated during baking to move more easily through the dough. Steam travels faster through air spaces than through dense dough particles or the gluten network. Therefore, large(er) air bubbles facilitate **more efficient steam movement**.

2. More uniform distribution of steam and more efficient baking: When steam moves more freely through the dough, it is distributed more evenly, leading to quicker and more uniform baking of the crumb.

3. Moisture loss: As steam diffuses more efficiently, it is also easier for it **to escape**. This is why **dough with higher hydration tends to lose more moisture during baking**, as its open structure promotes steam "release."

4. Creating an insulation layer and slowing heat transfer: The more open the crumb structure, the harder it is for oven heat to penetrate the center of the dough. Air bubbles, filled with CO2 (a good insulator), create **a layer of "thermal insulation."** The larger the air bubbles, the more effective the insulation layer, causing heat to concentrate in the crust area, and slowing down the transfer of heat to the dough's interior.

In conclusion, **higher hydration** results in larger air bubbles during baking, which creates more "paths" for steam to travel within the dough and escape. This makes the baking process **more efficient**, improving crumb stabilization (through gelatinization and gluten coagulation) and moisture evaporation (as steam escapes from the dough).

On the other hand, **lower hydration** creates a more closed crumb structure, which makes it harder for steam to circulate and escape. This means that dough with lower hydration **can retain moisture more effectively**, as steam finds it harder to escape.

Illustration: The Effect of Hydration on Moisture Loss During Baking

The table below, sourced from this study, illustrates how hydration affects moisture loss during baking. The study examined, among other things, various hydration levels and the impact of different flours on moisture loss and the final weight of bread.

The table shows the percentage of weight lost by the dough during baking. For our purposes, we'll focus on the **Straight Dough** column (which represents direct dough, without a sponge preferment).

The observed weight loss is solely due to moisture evaporation, as this is the only way dough loses weight during baking.

TABLE IV
Percent Bread Yield Loss in Weight of Dough During the Entire Baking Process from Doughs of the Buhler-Milled Flours that Varied in Optimum Baking Absorption and Baked with Different Absorption Levels

Variety	Baking Absorption (%)	Loss, %	
		Straight Dough	Sponge and Dough
Butte 86	+4% (73.1)	17.23 a[a]	18.78 a
	+2% (71.1)	16.58 ab	17.98 b
	Optimum (69.1)	16.39 b	17.88 b
	−2% (67.1)	15.96 bc	17.52 b
	−4% (65.1)	15.37 c	16.88 b
SD		0.061–0.405	0.127–1.010
Len	+4% (70.3)	16.71 a[a]	16.97 a
	+2% (68.3)	16.53 ab	16.39 b
	Optimum (66.3)	15.80 b	16.40 b
	−2% (64.3)	15.69 b	16.04 b
	−4% (62.3)	14.55 c	16.12 b
SD		0.085–0.324	0.218–0.734
Marshall	+4% (61.9)	16.71 a[a]	13.62 a
	+2% (59.9)	16.71 a	13.91 a
	Optimum (57.9)	15.99 b	14.97 a
	−2% (55.9)	15.31 c	14.99 a
	−4% (53.9)	14.04 d	14.53 a
SD		0.186–0.480	0.255–0.397

[a] Values followed by different letters indicate statistically significant differences at the 5% level.

'Baking Absorption' = Dough Hydration

From the data, it is clear that **higher hydration** levels lead to **greater moisture loss** during baking, regardless of the flour type (Butte 86, Len, and Marshall).

Crumb Moisture Content After Baking

Now that we understand that dough with higher hydration loses more moisture during baking, let's explore how this affects the **final moisture content** of baked products.

The Difference Between Dough Hydration and Moisture Content

First, it's important to clarify that dough hydration is calculated using **baker's percentage**, which is based on the weight of the **flour** in the dough, not the **total weight of the dough**.

To determine the **moisture content** of the dough, whether before or after baking, we need to consider it in relation to the **total weight of the dough (or the baked product)**, rather than just the flour.

Let's take an example of a dough weighing 265 grams, consisting of 165 grams of flour and 100 grams of water. The **hydration** of this dough is **60%** (100/165).

To calculate the **moisture content**, we use the **total dough weight** (265 grams). In this case:
100 / 265 = 0.377, or **37.7%** moisture content.

The flour itself contains about 14% moisture. For simplicity, we will exclude this moisture and only focus on the water added during dough preparation.

Now, consider a dough of the same weight (265 grams), but with 155 grams of flour and 110 grams of water. The **hydration** here is **70%** (110/155).

To calculate the moisture content:
110 / 265 = 0.415, or **41.5%** moisture content.

Below is a table summarizing doughs at 60%, 70%, and 80% hydration and their respective water content:

Dough Hydration (% water relative to the weight of flour)	**Moisture Content Before Baking** (% water relative to the total weight of the dough)
60%	38%
70%	41.5%
80%	44.5%

Notice how, despite the relatively large difference in **hydration**, the **moisture content** difference is much smaller.

Additionally, note that this represents the moisture content of the dough **before baking**, and before it loses any additional moisture during baking. As we'll see in the next section, this difference doesn't always carry over to the moisture content **after** baking.

Typically, a thin-crust pizza will have between 26% and 30% moisture after baking, while a Neapolitan pizza typically has around 40% moisture. By comparison, a loaf of bread generally contains 38% to 40% moisture.

The difference in moisture content between pizza and bread comes down to their structure. Bread has more crumb, which retains moisture more effectively than the crust. Pizza, being flatter, has a larger crust-to-crumb ratio, leading to greater moisture loss from the crust and a lower overall moisture content.

Illustration: The Effect of Hydration on Final Moisture Content

The table below, sourced from this study, illustrates how hydration levels (68.2%, 71.2%, and 74.2%) and baking durations (21, 24, and 27 minutes) affect the moisture content in both the crust and crumb of bread. Moisture content was measured after one hour and after 24 hours of cooling.

The key data for our discussion is the moisture % measured after **one hour**. The table on the **right** shows the moisture content in the **crust**, while the table on the **left** shows the moisture content in the **crumb**.

The table also includes the value $\mathbf{a_w}$, which represents water activity. Water activity measures the amount of "free" water that is not chemically bound to other components in the dough. While not directly relevant to this discussion, a_w is a useful measure in food safety.

The a_w scale, ranging from 0 to 1, helps assess the potential for spoilage or bacterial growth. Foods with higher a_w values are more likely to support microbial growth and undergo undesired chemical reactions, making them more susceptible to spoilage.

TABLE III
Water Activity (a_w) and Moisture Content of Crumb in Bread from Doughs that Varied in Baking Absorption and Baking Time[a]

Baking Absorption and Baking Time	a_w		Moisture (%)	
	1 hr	24 hr	1 hr	24 hr
Baking absorption (%)				
Optimum (71.2)	0.957 b	0.960 a	44.03 b	43.92 b
−3% (68.2)	0.956 b	0.962 a	43.26 c	43.03 c
+3% (74.2)	0.969 a	0.966 a	45.27 a	45.30 a
Standard deviation	0.001−0.004	0.001−0.003	0.127−0.346	0.057−0.396
Baking time (min)				
24 (optimum)	0.957 b	0.960 a	44.03 a	44.92 a
21	0.969 a	0.966 a	44.28 a	44.40 a
27	0.965 ab	0.966 a	44.08 a	44.05 a
Standard deviation	0.001−0.004	0.002−0.003	0.127−0.240	0.049−0.269

[a]Values followed by different letters indicate statistically significant differences at the 5% level.

TABLE IV
Water Activity (a_w) and Moisture Content of Crust in Bread from Doughs that Varied in Baking Absorption and Baking Time[a]

Baking Absorption and Baking Time	a_w		Moisture (%)	
	1 hr	24 hr	1 hr	24 hr
Baking absorption (%)				
Optimum (71.2)	0.886 a	0.890 a	18.03 a	25.01 a
−3% (68.2)	0.878 b	0.886 a	17.78 a	24.24 a
+3% (74.2)	0.888 a	0.895 a	18.47 a	24.84 a
Standard deviation	0.001−0.004	0.000−0.005	0.240−0.629	0.877−1.138
Baking time (min)				
24 (optimum)	0.886 b	0.890 b	18.03 b	25.01 a
21	0.912 a	0.909 a	21.60 a	26.00 a
27	0.849 c	0.872 c	15.50 c	22.81 a
Standard deviation	0.002−0.009	0.000−0.009	0.240−0.933	1.053−1.379

[a]Values followed by different letters indicate statistically significant differences at the 5% level.

Summary of Findings:

- **Hydration's Impact on Crumb Moisture:** There is a 2% difference in final moisture content between doughs with 68% and 74% hydration.
- **Baking Time's Effect on Crumb Moisture:** Baking time had little to no effect on the crumb moisture content. Crumb moisture was nearly identical between bread baked for 21 minutes and bread baked for 27 minutes, despite a 30% increase in baking time.
- **Hydration's Effect on Crust Moisture:** Hydration had a marginal to nonexistent effect on crust moisture content.
- **Baking Time's Effect on Crust Moisture:** Baking time significantly impacts the moisture content of the crust.

- **Moisture Migration**: After 24 hours, the crust absorbed significant moisture due to a phenomenon called "moisture migration," where moisture moves from the more humid crumb to the drier crust.

These findings highlight several key principles in baking, some of which we have discussed (or will discuss) in this article, and some of which were covered in the article [From Dough to Pizza: A Deep Dive into the Processes Occurring Inside the Dough During Baking]:

- **Moisture Loss from the Crust:** The majority of the moisture loss during baking occurs from the crust.
- **Crust Drying:** The longer the baking time, the more the crust "dries out" and loses moisture.
- **Minimal Crumb Moisture Loss:** The crumb loses minimal moisture during baking.
- **Impact of Hydration on Final Moisture Content:** While initial hydration does affect the final moisture content of the baked product, the difference is less significant than often assumed. Doughs with higher hydration lose more moisture during baking due to the more efficient escape of steam.

Adjusting Hydration to the Baking Temperature

As a general rule, and all things being equal (same dough, baking conditions, and times), **the higher the dough hydration, the lower the required baking temperature.**

This is because higher baking temperatures cause the crust to form more quickly. Once the crust forms, the crumb stops losing moisture. If dough with very high hydration is baked at too high a temperature, excessive moisture may be "trapped" in the crumb, resulting in a crumb that is underbaked and too "wet."

However, this principle does not apply in reverse: **baking at a lower temperature does not require higher hydration.**

Baking dough with high hydration at a lower temperature allows more time for moisture to evaporate before the crust forms. Since lower hydration doughs have a denser structure that slows moisture loss, the crumb retains moisture more effectively, making higher hydration **unnecessary** when baking at lower temperatures.

There is a common misconception that for baking at relatively low temperatures in a home oven (up to 300°C / 570°F), it's necessary to "adjust" the hydration by **increasing** it, to "prevent the dough from drying out." **This advice is misguided**.

While higher hydration results in dough with a higher final moisture content (regardless of baking temperature), it does not make the dough more suitable for home oven baking, nor does it prevent the dough from "drying out" during baking. As a **very** general guideline:

- For baking temperatures **up to about 350°C (660°F)**, no adjustments to dough hydration or baking temperatures are necessary.
- For baking temperatures **higher than 350°C (660°F)**, the higher the hydration, the longer the dough needs to be baked at a lower temperature.

For example, a classic Neapolitan pizza is baked at temperatures above 350°C. Its dough typically has a hydration level of 58-62%, and is baked at over 400°C (750°F) for no more than 90 seconds. This hydration level allows the dough to be fully baked in a very short time.

In contrast, **modern Neapolitan pizza** ("Napoletana Contemporanea"), with a dough hydration of 75% or more, is typically baked at a lower temperature (around 350°C) for 2-3 minutes. This longer baking time allows the excess moisture in the crumb to evaporate properly.

Baking high-hydration **modern** Neapolitan pizza at the same high temperature as the **classic** version would **trap too much moisture** inside, resulting in an **undercooked crumb** (as can be seen in the picture below).

Therefore, when baking at high temperatures, as **dough hydration increases**, both the **baking temperature should decrease**, and the **baking time should be extended**,to ensure proper moisture evaporation.

The Relationship Between Hydration, Browning, and Crispiness

Hydration levels can greatly affect the browning and crispiness of the crust.

In general, **higher hydration** results in **faster browning** and a **crispier crust**, all other things being equal.

While we've already touched on the reasons behind this in earlier sections, let's revisit them specifically in the context of browning and crispiness:

1. Higher hydration dough creates a **more open crumb structure** during baking, with larger air cells (air bubbles).
2. These air bubbles form a **thermal "insulation layer"** between the crust and the crumb.
3. This insulation layer causes heat to **concentrate in the crust**, making it more difficult for heat to penetrate into the interior of the dough.
4. As a result, **the crust receives more heat** and reaches a higher temperature more quickly.
5. This leads to a crust that is **more browned, drier, and crispier**.

Impact on Digestibility and "Lightness"

A common misconception is that higher dough hydration makes the finished product more "healthy," "digestible" or "light," under the assumption that it contains more water and less flour (for the same dough weight).

Let's clarify this by considering a simple example:

Suppose we examine three doughs with different hydration levels: 60%, 70%, and 80%, each weighing **300 grams**, which is a typical weight of a standard-sized pizza dough.

Dough Hydration (Baker's %)	Flour (Grams)	Water (Grams)	Flour % (Relative to Dough Weight)
60%	188	112	62.6%
70%	176	124	58.6%
80%	167	133	55.6%

The difference in flour content between the 60% hydration dough and the 80% hydration dough is about **7%**, or around **20 grams** of flour. The difference between the 60% and the 70% hydration doughs amounts to about 10 grams.

Now, is a 20-gram difference in flour significant enough to make the dough "lighter" or more "digestible" when eating an *entire* pizza? I highly doubt it.

To put it simply, no – this small difference in flour does not impact the dough's digestibility or "lightness" in any meaningful way.

How Dough Hydration Impacts Eating Characteristics

As we've seen, hydration affects how dough bakes, but its impact on the **eating characteristics** of finished products – whether pizza, bread, or other baked goods – is often **misunderstood**.

Contrary to popular belief, hydration does not directly or significantly determine the softness, moistness, or tenderness of baked goods.

To clarify this, the following sections will examine the relationship between dough hydration and key eating characteristics of baked goods. Specifically, we will explore:

- How moisture content affects the perception of **softness and "juiciness"** in baked goods (and other foods).
- How **"airiness"** contributes to the **softness and tenderness** of baked goods.

Distinguishing Between Moisture, Wetness, and Softness

Moisture, which refers to the **water content** of a food, is a measurable characteristic determined by the amount of water present. In contrast, **wetness/juiciness** and **softness** are sensory qualities, that can't be chemically measured, and aren't directly tied to moisture content.

Moisture refers to the **water content** in food, including both **free water** (which can move freely) and **bound water** (which is chemically or physically attached to food components). The more water a food contains, especially **free water**, the more 'moist' it will feel.

For example, soup has a high water content, making it feel very moist, while crackers contain very little water, making them dry.

Crackers' low water content, particularly their minimal free water (typically up to 5%), allows them to stay fresh and retain their quality for an extended period without spoiling.

However, **wetness** doesn't always correlate with moisture. A food can feel "wet", even with low moisture content. For example, oil, which contains no water (100% fat), feels wet, as does peanut butter, despite its low moisture content.

Similarly, **softness** is not directly related to moisture. A food can be soft with minimal moisture, and vice versa – a food can have high moisture, yet feel firm or rigid. For instance, **marshmallows** are very soft, but have relatively low moisture content (around 20%), while **apples**, with about 85% water, are not soft at all.

In conclusion, **there is no direct relationship between a dough's moisture content** (determined by its hydration level) **and the perceived softness or wetness of the finished baked product**.

Dough with higher hydration or moisture content will not necessarily feel "wetter" or "juicier" when eaten. Hydration and moisture levels in the dough are just one factor – albeit a relatively **minor** one – among many that influence the perception of softness or juiciness in the final baked product.

Variables such as dough composition, baking temperature, baking time, and the quality of fermentation play a much more significant role in determining the eating characteristics of the product. For example, a dough with 60% hydration and 2% oil will be perceived as "wetter" and "juicier" than a dough with 80% hydration and no oil, assuming all other factors are equal.

Interestingly, baked goods that have become hard and stiff over time – commonly perceived as "dry" or "spoiled" – actually retain **similar moisture levels** to their fresh counterparts. In standard storage conditions, where the environment is not very dry, **baked products lose very little moisture during storage**, maintaining similar water content whether they are "fresh" or "stale."

The hardening of baked goods **is not caused by a loss of moisture or "drying out**," but by a process called **starch retrogradation**. During retrogradation, starch molecules recrystallize and stiffen, leading to the product's hardening. This phenomenon affects any baked good containing flour (starch), regardless of moisture content, although various factors can influence how quickly this process occurs.

This example highlights how variables unrelated to moisture content can dramatically influence the perception of juiciness, wetness or softness. Both a fresh baked product and its hardened counterpart may contain similar moisture levels, yet their eating characteristics can feel entirely different.

The Relationship Between Airiness, Softness, and Tenderness

As we've seen, higher hydration leads to a more airy and open crumb structure. This often results in a crumb that feels softer **to the touch**, because it is less "compressed", and creates less resistance when pressure is applied.

However, it's important not to confuse "airiness" or "softness" with "tenderness."

Understanding the Difference Between Softness and Tenderness

- **Tenderness** refers to a melt-in-the-mouth quality. A tender food **breaks down easily when chewed**, requiring minimal effort. This is in contrast to foods with chewy or tough characteristics, which demand more effort to bite and chew.
- **Softness**, on the other hand, relates to how much resistance a food offers **when pressure is applied**. A **soft** food may resist less, but is not necessarily tender or easy to chew.

A common misconception in pizza-making is assuming an airy crumb – achieved through higher hydration – is also tender. In reality, **these are distinct properties that do not directly influence each other**.

Examples Highlighting the Difference:

- **Soft, but not Tender**: Marshmallows are soft, but dense and chewy; Gummy bears are also soft, yet require significant chewing effort.
- **Tender, Yet Not Soft to the Touch**: A properly prepared fish fillet may feel firm and springy when touched, but it is tender and easy to chew once eaten.
- **Dense, Yet Soft and Tender**: Baked goods like brioche and danishes often have a **dense, non-airy crumb structure**, yet are **soft, fluffy, and tender** when eaten. Their tenderness comes from their **fat content**, not high hydration. Despite their relatively low hydration (50-60%), they are among the softest and most tender baked goods.
- **Airy, Yet Tough**: Artisan sourdough breads often feature a light, open crumb structure, but tend to be **chewier** than bread made with baker's yeast, even when all other factors are the same.

The chewiness of sourdough breads is due to the high acidity in the dough, which strengthens the gluten, making it more elastic and chewy (for more on this, see the article [Dough Elasticity and Extensibility: Understanding the Two Most Important Properties in Pizza Dough]).

High Hydration Is Not the Solution to Tenderness

While high hydration can produce a baked product that feels **softer to the touch**, it does not necessarily result in a **tender, easier to chew, melt-in-the-mouth product**.

If the goal is a baked product that is **soft and tender to eat**, requiring little chewing effort, high hydration is NOT the answer. Factors like dough composition (specifically fat content), proper fermentation, kneading method and baking techniques, play a much larger role in achieving tenderness.

A common question in pizza making is: "*How can I make my pizza more tender and less chewy or tough?*" (often using "soft" instead of "tender" and "rubbery" instead of "chewy" or "tough").

Many mistakenly believe that increasing hydration will help. However, achieving a tender pizza depends on factors like fermentation, fat content, and gluten development, rather than simply adding more water to the dough.

Conclusion: Does Higher Hydration Mean "Better" Pizza?

The short answer is: **no, higher hydration does not necessarily result in a "better" product**, whether pizza or bread, and **making a high-quality product does not require high hydration**. In fact, increasing hydration might make the dough harder to handle, which could negatively impact the final product.

As we've discussed, hydration affects how the dough bakes and its final structure. Higher hydration **allows** for:

- A baked product with a **larger volume**.
- A baked product with a **more open, airy crumb structure** and **larger "air bubbles."**
- A baked product with a slightly higher final moisture content (which has a marginal effect on actual eating characteristics).
- A product that may be **softer to the touch** due to the more open structure, but not necessarily more tender.
- Faster, more intense browning.
- A crispier crust (for pizza).

However, increasing hydration **will not**:

- Make the crumb more tender.

- Make the crumb more “wet” or “juicy.”
- Make the baked product “healthier,” “lighter,” or “more digestible.”
- Prevent the crumb from “drying out” when baking at lower temperatures.
- Enhance crispiness.

In summary, increasing hydration leads to two main practical outcomes:

1. A more open, airy crumb structure.
2. A larger volume.

Beyond that, higher hydration offers no inherent improvement, and is only beneficial if you aim for the outcomes mentioned above. Otherwise, it can make the dough harder to work with, potentially compromising the overall quality of the finished product. **Increasing hydration will not automatically improve your pizza, bread, or any other baked good**.

While higher hydration has its advantages and specific applications (for example, it’s essential in wholemeal doughs), it’s important to understand that higher hydration alone won’t make your product better, nor is it an indicator of a baker’s skill level.

Personal Note on High Hydration Doughs

Personally, I’m not a fan of high hydration for pizza dough. While high hydration can create the large “air bubbles” that look appealing in photos, the effect on the actual eating quality of the pizza is **minimal**, if not **nonexistent**.

In today’s world of social media, where visuals often dictate trends, it’s not surprising that many people prioritize creating pizzas that photograph well, rather than focusing on how they eat.

The allure of large air bubbles has become a **misleading indicator of quality**, both in pizza and the skill of the baker. While this aesthetic may be visually appealing, it doesn’t necessarily lead to a better eating experience – often, it’s quite the opposite.

This trend extends beyond pizza into bread baking, where the goal of achieving a crumb filled with the largest possible “air bubbles” has become fashionable.

In some cases, up to 80% of the crumb is holes (“air bubbles”), leaving only 20% as actual bread. Such a loaf is impractical – it’s not spread-friendly, doesn’t make good sandwiches, and frankly, it’s just not enjoyable to eat (trust me, I’ve tried).

This shift in priorities illustrates a growing trend in the baking world (and food in general): focusing on what looks good for Instagram, rather than what's enjoyable to eat. Unfortunately, many bakers are falling into this cycle, but if you've read this far, you have the opportunity to break free from it.

Ultimately, if your goal is to create a pizza with **great eating qualities**, rather than one that **just looks good on Instagram**, higher hydration is not necessarily the key. In fact, for most applications, [higher] dough hydration has a negligible impact on the overall eating qualities of pizza (or bread).

Typical Hydration Levels for Various Pizza Styles and Other Baked Products

Below are general and "traditional" hydration levels for various types of pizzas and other baked goods.

Pizzas

New York Style Pizza: 57-62%

New Haven Pizza: 65-70%

Classic Neapolitan Pizza: 58-62%

Modern Neapolitan Pizza ("Napoletana Contemporanea"): 70-80%

Classic Italian Pizza (Pizza al Piatto / Pizza Tonda): 55-60%

Pizza Tonda Romana: 55-58%

Detroit Style Pizza: 65-70%

Sicilian Pizza: 57-62%

Pizza al Taglio Romana (Classic): 60%

Pizza al Taglio Romana (Modern): 75-80%

Pizza Pala: 70-80%

Pizza Pinsa: 70-80%

Industrial-Style Pizza (Pizza Hut, Domino's, Papa John's, etc.): 55-58%

Chicago Deep Dish Pizza: 45-50%

Cracker-Style Pizza: 35-50%

Bar Pizza (Bar Pie) / Tavern-Style Pizza / Midwest Style Pizza: 50-60%

Other Baked Products

Standard Bread and Buns: 55-60%

Artisanal Breads: 60-90% (Varies)

Challah: 55-60% (Including water from eggs and butter)

Brioche: 50-60% (Including water from butter)

Baguette: 60-65%

Ciabatta and Focaccia: 60-80% (Varies)

Hoagie / Sub Roll: 58-62%

Various Flatbreads (such as Pitta Bread): 55-58%

New York Bagle: 57-60%

Burger and Hotdog Buns: 57-60%

Shortcrust and Puff Pastry: 25-35% (*Not* including water from butter)

Pasta Dough: ~40% (Water that comes only from the eggs)

Bao Buns (and other steamed buns): 50-55%

Babka / Krantz: 50-58% (Including water from eggs and butter)

Croissant: 50-60% (Not including water from butter used for lamination)

Danishes: 50-60% (Including water from eggs and butter, not including water from butter used for lamination)

Key Takeaways

- **The Core Definition:**
  Hydration is the weight of water relative to flour, expressed in Baker's Percentage. It dictates the dough's consistency: lower hydration creates a stiff, elastic dough, while higher hydration results in a soft, "liquid," and extensible dough.

- **Quality vs. Trends:**
  Higher hydration does **not** equal a better pizza. The pursuit of extreme hydration is largely a social media trend; a pizza's quality is defined by fermentation, handling, and bake, not the sheer volume of water in the mix.
- **Oven Spring and Crumb:**
  Higher hydration facilitates greater expansion during baking (**oven spring**). This leads to a more "open" and airy crumb structure ("alveoli" or "air bubbles"), which is the primary reason to increase hydration.
- **Hydration & Crispiness:**
  contrary to the common belief, **Higher** hydration leads to **enhanced crispiness/browning** – not the other way around. Higher hydration leads to a more airy crust, which becomes more **isolated from heat** – this insulation allows the outer shell to brown and crisp faster.
- **The Moisture Paradox:**
  Hydration has a negligible effect on the moisture of the final baked crust. A properly baked pizza will have roughly the **same moisture content post-bake** regardless of the initial hydration; more water does not result in a "moister" pizza.
- **The Lightness/Digestibility Myth:**
  Dough hydration has a negligible impact on digestibility or the "lightness" of the crust. The difference in flour volume between standard and high hydration is too small to affect how "heavy" the pizza feels in the stomach.
- **Softness and Chew:**
  Higher hydration does **not** make a crust softer, more tender, or less chewy. Tenderness is dictated by other factors like the flour used, bake time, fats used, dough handling and fermentation – hydration is not a primary factor in the final "bite" texture.
- **The Temperature-Evaporation Relationship:**
  Higher hydration requires **lower** temperatures and **longer** bake times to allow excess moisture to evaporate before the crust sets; baking wet dough at extreme heat, like a modern Neapolitan pizza (Napoletana contemporanea) "traps" moisture inside, resulting in a gummy, undercooked crumb. However, **this principle does not apply in reverse**: baking at a lower temperature (such as in a home oven) does not necessitate adding more wate.
- **The Home Oven Misconception:**
  Contrary to popular belief, baking at lower temperatures (home ovens up to 300°C/570°F) does **not** require higher hydration. Because lower-hydration dough is denser (due to less oven spring), it actually retains moisture *better*. High hydration is not a tool to prevent the dough from "drying out."

- **Effective Hydration:**
  To find your “true” hydration, you must account for the water content in milk (90% water), eggs (90% water), butter (15% water), etc. Conversely, oil is 100% fat and contains no water – it does not hydrate the flour or contribute to gluten development and biochemical fermentation processes, therefore, it should not be calculated as dough hydration.
- **Hydration & Fermentation Rate:**
  While water is necessary for yeast activity, increasing hydration beyond 60% does not inherently speed up fermentation. The “faster rise” seen in wet doughs is often a physical illusion caused by lower structural resistance to gas expansion.
- **The Practical Range:**
  For the vast majority of pizza styles, a hydration of **55–65%** is optimal. You should only exceed this range if you are specifically targeting a more open crumb structure or very specific crust characteristics.